

Systems modelling for climate adaptation | Climate risk | Environmental data science

Research interests: climate adaptation and resilience; complex systems; climate-risk modelling; spatio-temporal dynamics; agent-based and network models.

CURRENT ENGAGEMENT

Climate Data Consultant (Part-time)

Sep 2026–Jan 2027

Khushi Baby | Climate-health vulnerability and environmental data methodology

- Standardizing the methodology for environmental variables used in the Climate Health Vulnerability Index (CHVI).
- Assessing appropriate spatio-temporal aggregation scales for input data and documenting scale-dependent trade-offs for vulnerability analysis.

EDUCATION

Indian Institute of Science Education and Research Pune (IISER Pune)

2021–2026

Integrated BS-MS in Science | *Applied Mathematics and Statistics* | CGPA: 8.8/10

- BS-MS thesis: From Grid Cells to Policy in Heat-Stress Analysis; supervised by Joy M. Monteiro (IISER Pune) and Deepak Malghan (IIM Bangalore).
- Institute Best MS Thesis Award (Data Science).

WORKING PAPERS & ONGOING RESEARCH

Mind the (Jensen) Gap: Climate and Health using Gridded Data

Mar 2026

Vatsal Maurya, Joy M. Monteiro, Anish Sugathan, Deepak Malghan | *IIMB Working Paper No. 732/2026*

- Develops a multiscale framework for nonlinear environmental metrics and threshold exceedances, showing how spatio-temporal aggregation can produce systematic, spatially clustered and threshold-sensitive bias.

Heat-stress metrics and the timing of peak season in India

Ongoing

Manuscript in preparation

- Compare the annual contiguous 90-day windows that maximize air temperature, WBGT, UTCI and Extended Heat Index using hourly ERA5 data for 1980–2024; preliminary results show substantial timing disagreement despite strong hour-to-hour metric correlation.

IMD heat-wave criteria through uncompensable heat strain

Ongoing

Ongoing manuscript project

- Contributed the initial partitional-calorimetry implementation and exploratory analysis using HadISD station records for India, Pakistan and Bangladesh.
- Ongoing collaboration evaluates how temperature-based IMD heat-wave declarations align with physiological heat risk by comparing uncompensable heat-strain hours on declared and non-declared days; manuscript in preparation.

RESEARCH EXPERIENCE

BS-MS Thesis Research — From Grid Cells to Policy in Heat-Stress Analysis

2025–2026

IISER Pune & Indian Institute of Management Bangalore | *Supervisors: Joy M. Monteiro and Deepak Malghan*

- Built and analysed global hourly ERA5/ERA5-Land heat-stress data at native resolution, using parallel and chunked workflows to compute UTCI, WBGT and meteorological intermediates; the ERA5 archive used in the thesis spans 1980–2024 and ~11–12 TB.
- Developed a multiscale framework for translating grid-level climate hazards to administrative decision scales, quantifying Jensen gaps introduced by nonlinear aggregation and thresholding.
- Modelled long-run changes in the temporal organisation of heat risk across India, including shifts in hot-season onset, duration, event persistence and recovery breaks over 45 years.
- Fit spatially explicit gridpoint logistic models of changing heat-event regimes, estimating evolving contributions of temperature, humidity, radiation, wind and daytime conditions.

ADDITIONAL RESEARCH EXPERIENCE

- Graph-theoretic analysis of road networks to identify critical road segments**Jun 2024–Jan 2025
- Online project, IIASA Vienna | Advisors: Jose Moran and Celian Colon*
- Modelled road infrastructure as a network system and applied graph-theoretic and simulation-based methods in NetworkX to identify critical and vulnerable segments.
- Human body energy model for physiologically dangerous heat events**Aug 2023–Dec 2024
- IISER Pune | Advisor: Joy M. Monteiro*
- Implemented a partitional-calorimetry model in Python and evaluated it on station-level meteorological data; this work provided the computational foundation for the ongoing IMD heat-wave criteria project.
- Agent-based models of economic mobility and inequality**May–Jul 2022
- ETC Lab, Jawaharlal Nehru University | Advisor: Anirban Chakraborti*
- Implemented Schelling and kinetic wealth-exchange models, including hybrid variants, to study how local interaction rules generate system-level patterns of inequality and mobility.

TECHNICAL SKILLS

Programming: Python (primary), C++, Java

Scientific / statistical computing: statsmodels, PySAL/esda, pymannkendall, NumPy/SymPy, NetworkX

Geospatial / climate data: ERA5, ERA5-Land, HadISD, QGIS, NetCDF, Zarr; parallel and chunked large-scale data workflows

Modelling: spatio-temporal statistical modelling, logistic regression, network analysis, agent-based modelling, partitional calorimetry

Languages: English (professional), Hindi (native), Kannada (elementary)

SELECTED AWARDS & SCHOLARSHIPS

- KVPY-SX Academic Fellowship, 2021–2026.
- Institute Best MS Thesis Award (Data Science), IISER Pune.

OTHER LEADERSHIP

- Sponsorship Coordinator — Karavaan 2022**Aug–Nov 2022
- IISER Pune*
- Led sponsorship outreach for the institute’s annual cultural festival and coordinated a team across departments.